

Machine Learning

Expanded Knowledge Base Document for RAG / Document Question-Answering

1. Introduction

Machine learning is a branch of artificial intelligence in which algorithms learn patterns from data and use those patterns to make predictions or decisions. Instead of writing every rule manually, engineers provide examples and an objective for learning. The learned model can then be applied to new observations.

2. Supervised Learning

Supervised learning uses labeled examples. Classification predicts categories such as spam or not spam, while regression predicts numerical values such as price. A supervised dataset normally contains input features and a target. During training, the model adjusts its parameters to reduce an objective or loss, and later its predictions are evaluated on data that was not used to fit the model.

3. Unsupervised Learning

Unsupervised learning works without target labels and can discover groups, structures, or lower-dimensional representations. Clustering is commonly used to group observations based on similarity. Dimensionality-reduction methods can represent data using fewer variables while attempting to preserve useful structure.

4. Machine Learning Workflow

A typical workflow includes collecting data, cleaning it, exploring it, selecting features, splitting data into training and evaluation sets, training a model, tuning hyperparameters, and measuring performance. Each stage affects the quality of the final system. Data leakage should be avoided so that information from evaluation data does not improperly influence model fitting.

5. Features and Preprocessing

Features are the variables used by a model to make predictions. Preprocessing may include handling missing values, encoding categories, scaling numerical variables, removing irrelevant information, and transforming variables. Feature engineering creates useful representations from available information, while feature selection attempts to retain informative variables and reduce unnecessary complexity.

6. Model Evaluation

Model evaluation should match the task. Classification may use accuracy, precision, recall, F1 score, or ROC-AUC. Regression may use mean absolute error, mean squared error, or R-squared. A model should be evaluated on data that was not used to fit its parameters. Cross-validation can provide a more reliable estimate of generalization by repeatedly training and evaluating across different splits.

7. Hyperparameter Tuning

Hyperparameters are settings chosen before or around model training, such as tree depth, regularization strength, learning rate, or the number of neighbors. Grid search and randomized search

are common approaches. Tuning should be performed using training data and an appropriate validation strategy rather than repeatedly optimizing directly on the final test set.

8. Ensemble Learning

Ensemble learning combines multiple models or multiple predictions to obtain a stronger overall system. Bagging methods train models on varied samples and combine their predictions. Boosting builds models sequentially so that later models focus on errors made by earlier models. Ensembles can improve predictive performance but may increase computational or interpretability requirements.

9. Overfitting and Underfitting

Overfitting occurs when a model learns training-specific patterns that do not generalize well.

Underfitting occurs when the model is too limited to capture useful relationships in the data.

Regularization, appropriate model complexity, cross-validation, more representative data, and careful feature selection can help address these problems.

10. Key Takeaways

Machine learning is a complete workflow rather than simply choosing an algorithm. Reliable results depend on representative data, correct preprocessing, appropriate validation, suitable metrics, careful tuning, and evaluation on unseen data.